

Reflected-Packet Interactions in the Semilocal Weil Form

An Exact Zero Expansion and Unconditional Infinite Oscillation

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Abstract

We freeze one explicit real, even, compactly supported smooth packet and compute the interaction of two widely separated reflected translates in the complete semilocal Weil preform. The pole and all-prime-power sectors combine into a Stieltjes pairing against $d\psi - dx$. Its canonical bilateral Laplace transform is

$$L_h(s) \left\{ \frac{\zeta'}{\zeta} \left(s + \frac{1}{2} \right) + \frac{1}{s - \frac{1}{2}} \right\}.$$

A contour displacement gives an absolutely convergent expansion over all nontrivial and trivial zeros. The archimedean term and the remaining pole term cancel the complete trivial-zero series, leaving a zero-only expansion for the full reflected cross interaction. The packet coefficient vanishes exactly on the lattice $4\pi i\mathbb{Z} \setminus \{0\}$. Conrey's unconditional positive-proportion theorem therefore supplies infinitely many visible critical-line poles. A Laplace-transform form of Landau's theorem then excludes either eventual sign. Consequently both the arithmetic interaction and the even-minus-odd energy splitting take both strict signs arbitrarily far out.

Scope. This is not a proof of the Riemann hypothesis. It is a fixed-packet theorem. It does not identify, approximate, or determine the parity of an actual semilocal ground state. Numerical computations in the repository are diagnostics only and are not premises of the analytic proof.

Contents

1	Introduction and main result	2
2	Frozen packet and semilocal normalization	3
2.1	Transform and polarization conventions	3
2.2	Complete semilocal Weil preform	4
2.3	The dyadic infinite-convolution packet	4
3	Polarization and exact prime–pole interaction	5
4	Canonical transform and exact zero expansion	6
4.1	The base-1 extension	6
4.2	Bilateral Laplace transform	7
4.3	Contour displacement	7
4.4	Archimedean cancellation	8

5	Visible zero coefficients and infinite oscillation	8
5.1	Coefficient visibility	8
5.2	A Laplace-transform Landau lemma	9
5.3	Oscillation of the arithmetic interaction	9
5.4	Oscillation of the complete splitting	10
6	Numerical diagnostics and finite certificates	10
6.1	Implementation A: direct prime-power side	10
6.2	Implementation B: independent Fourier and zero sides	10
7	Logical scope and independent audit	11

1 Introduction and main result

The Weil explicit formula balances four sectors: an archimedean distribution, two residual pole terms, the full prime-power measure, and a spectral sum over the nontrivial zeros. We study a particularly rigid test: the interaction of two reflected translates of one fixed packet. The purpose is arithmetic. No spectral ground-state or variational assertion is made.

Let $h \in C_c^\infty(\mathbb{R})$ be the packet constructed in Section 2. For $y > 1/2$, set

$$h_y^R(x) = h(x - y), \quad h_y^L(x) = h(x + y),$$

and

$$h_y^\pm = \frac{h_y^R \pm h_y^L}{\sqrt{2}}.$$

The supports of h_y^R and h_y^L are disjoint, so both $h_y^\pm$ are normalized. The energy splitting is

$$\Delta_h(y) = \mathfrak{q}[h_y^+] - \mathfrak{q}[h_y^-].$$

Write

$$K_h = h * h, \quad L_h(w) = \int_{-1}^1 K_h(r) e^{wr} dr$$

and define the all-prime-power error measure

$$dE(x) = \sum_{n \geq 2} \Lambda(n) \delta_n - dx.$$

The centered arithmetic interaction is

$$\mathcal{I}_h(y) = - \int_0^\infty x^{-1/2} K_h(2y - \log x) dE(x).$$

Theorem 1.1 (Exact expansion and unconditional oscillation). *For every $y > 1/2$,*

$$\begin{aligned} \mathcal{I}_h(y) = & \sum_{\rho} m_{\rho} L_h \left(\rho - \frac{1}{2} \right) e^{2(\rho-1/2)y} \\ & + \sum_{n=1}^{\infty} L_h \left(2n + \frac{1}{2} \right) e^{-(4n+1)y}, \end{aligned} \tag{1}$$

where ρ ranges over the distinct nontrivial zeros of ζ and m_{ρ} is the multiplicity. Both series converge absolutely, and the nontrivial-zero series may be differentiated term by term any fixed number of times locally uniformly in y .

The archimedean and residual pole terms cancel the second series exactly. Consequently

$$\mathfrak{q}(h_y^R, h_y^L) = \sum_{\rho} m_{\rho} L_h \left(\rho - \frac{1}{2} \right) e^{2(\rho-1/2)y}, \quad (2)$$

and

$$\Delta_h(y) = 2 \sum_{\rho} m_{\rho} L_h \left(\rho - \frac{1}{2} \right) e^{2(\rho-1/2)y}. \quad (3)$$

The coefficient at ρ is

$$A_h(\rho) = m_{\rho} L_h \left(\rho - \frac{1}{2} \right),$$

and it vanishes exactly if

$$\rho = \frac{1}{2} + 4\pi i k, \quad k \in \mathbb{Z} \setminus \{0\}.$$

Unconditionally, for every $Y > 0$, there exist $y_+, y_-, u_+, u_- > Y$ such that

$$\mathcal{I}_h(y_+) > 0, \quad \mathcal{I}_h(y_-) < 0,$$

and

$$\Delta_h(u_+) > 0, \quad \Delta_h(u_-) < 0.$$

Thus each function has infinitely many sign-crossing intervals.

The sign convention has a direct but limited interpretation: $\Delta_h(y) < 0$ means that the even packet has lower energy than the odd packet at that separation, while $\Delta_h(y) > 0$ means the reverse. Theorem 1.1 therefore proves that neither parity is eventually preferred for this one packet.

The proof has four independent components. First, exact polarization fixes the factor 2. Second, the pole and prime-power main terms are combined before estimation. Third, a contour displacement produces (1), including every trivial zero. Finally, an exact packet-transform zero set and Conrey's theorem produce visible nonreal poles; Landau's theorem then rules out either eventual sign.

2 Frozen packet and semilocal normalization

2.1 Transform and polarization conventions

The inner product on $L^2(\mathbb{R}, dx)$ is antilinear in the first variable:

$$\langle \Phi, \Psi \rangle = \int_{\mathbb{R}} \overline{\Phi(x)} \Psi(x) dx.$$

Our Fourier transform and inverse normalization are

$$\widehat{\Phi}(z) = \int_{\mathbb{R}} \Phi(x) e^{-izx} dx, \quad \langle \Phi, \Psi \rangle = \frac{1}{2\pi} \int_{\mathbb{R}} \overline{\widehat{\Phi}(t)} \widehat{\Psi}(t) dt.$$

Translation $T_y \Phi(x) = \Phi(x - y)$ obeys

$$\widehat{T_y \Phi}(t) = e^{-iyt} \widehat{\Phi}(t).$$

For multiplicative tests we use

$$\widetilde{f}(s) = \int_0^{\infty} f(u) u^{s-1} du.$$

For $\Phi, \Psi \in C_c^{\infty}(\mathbb{R})$, define

$$C_{\Phi, \Psi}(r) = \int_{\mathbb{R}} \overline{\Phi(x - r)} \Psi(x) dx.$$

2.2 Complete semilocal Weil preform

The source normalization reconstructed from [1, 2] is

$$\begin{aligned} q(\Phi, \Psi) = & \frac{1}{2\pi} \int_{\mathbb{R}} m_{\infty}(t) \overline{\widehat{\Phi}(t)} \widehat{\Psi}(t) dt \\ & + \overline{\widehat{\Phi}(-i/2)} \widehat{\Psi}(i/2) + \overline{\widehat{\Phi}(i/2)} \widehat{\Psi}(-i/2) \\ & - \sum_{\substack{p \text{ prime} \\ m \geq 1}} \frac{\log p}{p^{m/2}} \{C_{\Phi, \Psi}(m \log p) + C_{\Phi, \Psi}(-m \log p)\}, \end{aligned} \quad (4)$$

where

$$m_{\infty}(t) = \Re \frac{\Gamma'}{\Gamma} \left(\frac{1}{4} + \frac{it}{2} \right) - \log \pi.$$

Every prime power p^m , rather than only every prime, appears with exact coefficient

$$-(\log p)p^{-m/2} = -\Lambda(p^m)(p^m)^{-1/2}.$$

For a finite semilocal interval $[-\log \lambda, \log \lambda]$, the cutoff is

$$p^m \leq \lambda^2.$$

The endpoint is inclusive and carries no half weight.

The equivalent real-place distribution is

$$\begin{aligned} W_{\mathbb{R}}(C) = & (\log 4\pi + \gamma_E)C(0) \\ & + \int_0^{\infty} \frac{e^{r/2} \{C(r) + C(-r)\} - 2C(0)}{e^r - e^{-r}} dr, \end{aligned} \quad (5)$$

and the archimedean contribution to (4) is $-W_{\mathbb{R}}$. This fixes the scalar, subtraction, and sign.

2.3 The dyadic infinite-convolution packet

For $a_n = 2^{-n-1}$, $n \geq 1$, put

$$b_{a_n}(x) = \frac{1}{2a_n} \mathbf{1}_{[-a_n, a_n]}(x)$$

and define

$$g = *_{n \geq 1} b_{a_n}, \quad h = \frac{g}{c_g}, \quad c_g = \|g\|_2.$$

Since $\sum_{n \geq 1} a_n = 1/2$, the infinite convolution is the density of an absolutely convergent sum of independent compactly supported random variables. Moreover

$$g \in C_c^{\infty}(\mathbb{R}), \quad \text{supp } g = [-1/2, 1/2],$$

and g is real, even, nonnegative, and strictly positive in the interior. Smoothness follows from the super-polynomial Fourier decay obtained by bounding any fixed number of sinc factors.

The transform is

$$\widehat{h}(z) = c_g^{-1} \prod_{n \geq 1} \frac{\sin(a_n z)}{a_n z}.$$

The product converges locally uniformly, and

$$Z(\widehat{h}) = 4\pi\mathbb{Z} \setminus \{0\}.$$

Because h is even,

$$K_h(r) = \int_{\mathbb{R}} h(x-r)h(x) dx = (h * h)(r).$$

It is real, even, nonnegative, smooth, supported exactly on $[-1, 1]$, positive on $(-1, 1)$, and $K_h(0) = 1$. Its bilateral Laplace transform is

$$L_h(w) = c_g^{-2} \prod_{n \geq 1} \left(\frac{\sinh(a_n w)}{a_n w} \right)^2. \quad (6)$$

This is entire and even. Since

$$\frac{\sinh(a_n w)}{a_n w} = 1 + O(a_n^2)$$

uniformly on compact w -sets and $\sum a_n^2 < \infty$, the infinite tail has no extra zeros. Hence

$$Z(L_h) = 4\pi i\mathbb{Z} \setminus \{0\}. \quad (7)$$

At $4\pi i k \neq 0$, the multiplicity is

$$2(v_2(|k|) + 1).$$

3 Polarization and exact prime–pole interaction

For $y > 1/2$, the two translates have disjoint support. Hermitian polarization, with the first form variable antilinear, gives

$$\begin{aligned} \mathfrak{q}[h_y^+] &= \frac{1}{2} \{ \mathfrak{q}[h_y^R] + \mathfrak{q}[h_y^L] \} + \Re \mathfrak{q}(h_y^R, h_y^L), \\ \mathfrak{q}[h_y^-] &= \frac{1}{2} \{ \mathfrak{q}[h_y^R] + \mathfrak{q}[h_y^L] \} - \Re \mathfrak{q}(h_y^R, h_y^L). \end{aligned}$$

Inversion symmetry makes the cross term real. Thus

$$\boxed{\Delta_h(y) = 2\mathfrak{q}(h_y^R, h_y^L)}. \quad (8)$$

Let

$$M_h = \widehat{h}(i/2) = \widehat{h}(-i/2) = \int_{\mathbb{R}} h(x) e^{x/2} dx > 0.$$

The two pole terms in (4) contribute exactly

$$M_h^2(e^y + e^{-y}). \quad (9)$$

The prime-power cross term is

$$- \sum_{p, m \geq 1} \frac{\log p}{p^{m/2}} K_h(2y - m \log p). \quad (10)$$

Because K_h is supported on $[-1, 1]$, only

$$2y - 1 < m \log p < 2y + 1$$

contributes. Equality is zero because $K_h(\pm 1) = 0$. This is a support fact, not an endpoint half-weight convention.

Define the full Chebyshev measure

$$d\Psi = \sum_{n \geq 2} \Lambda(n) \delta_n, \quad \psi(X) = \sum_{n \leq X} \Lambda(n),$$

and

$$dE = d\Psi - dx.$$

This is the prime-power error distribution associated with $\psi(X) - X$, not the primes-only distribution associated with $\theta(X) - X$.

The continuous term can be evaluated without approximation:

$$\begin{aligned} \int_0^\infty x^{-1/2} K_h(2y - \log x) dx &= e^y \int_{-1}^1 K_h(r) e^{-r/2} dr \\ &= e^y M_h^2. \end{aligned} \tag{11}$$

Therefore

$$\mathcal{I}_h(y) = e^y M_h^2 - \sum_{n \geq 2} \frac{\Lambda(n)}{\sqrt{n}} K_h(2y - \log n), \tag{12}$$

and the exact cross interaction is

$$\mathfrak{q}(h_y^R, h_y^L) = A_h^\infty(y) + e^{-y} M_h^2 + \mathcal{I}_h(y), \tag{13}$$

where

$$A_h^\infty(y) = \frac{1}{2\pi} \int_{\mathbb{R}} m_\infty(t) e^{2iyt} |\widehat{h}(t)|^2 dt.$$

Equation (13) is the exact prime-pole cancellation: the $e^y M_h^2$ pole main term has been combined with the continuous part of dE before any estimate is made.

For a packet realized inside $[-\log \lambda, \log \lambda]$, the geometric condition is

$$y + \frac{1}{2} \leq \log \lambda.$$

Every contributing prime power then automatically satisfies the inclusive semilocal cutoff $p^m \leq \lambda^2$.

4 Canonical transform and exact zero expansion

4.1 The base-1 extension

The uncut Mellin transform of dE is not defined in a right half-plane, because $dE = -dx$ on $(0, 1)$. Introduce only for the transform

$$J_h(t) = - \int_{[1, \infty)} x^{-1/2} K_h(t - \log x) dE(x).$$

If

$$E_0(x) = \psi(x) - x, \quad E_1(x) = \psi(x) - (x - 1) \quad (x \geq 1),$$

then $E_0(1) = -1$, $E_1(1) = 0$, and distributionally

$$d(\mathbf{1}_{[1, \infty)} E_0) = \mathbf{1}_{[1, \infty)} dE - \delta_1,$$

whereas

$$d(\mathbf{1}_{[1,\infty)}E_1) = \mathbf{1}_{[1,\infty)}dE.$$

Thus J_h uses the endpoint-normalized primitive E_1 and contains no spurious $K_h(t)$ boundary term. It satisfies

$$J_h(t) = 0 \quad (t \leq -1), \quad J_h(t) = \mathcal{I}_h(t/2) \quad (t > 1).$$

4.2 Bilateral Laplace transform

For $\Re s > 1/2$, absolute Euler-product convergence and Tonelli's theorem give

$$\begin{aligned} \mathcal{B}_h(s) &= \int_{\mathbb{R}} J_h(t) e^{-st} dt \\ &= L_h(s) \left\{ \frac{\zeta'}{\zeta} \left(s + \frac{1}{2} \right) + \frac{1}{s - \frac{1}{2}} \right\}. \end{aligned} \tag{14}$$

At $s = 1/2$,

$$\frac{\zeta'}{\zeta} \left(s + \frac{1}{2} \right) = -\frac{1}{s - 1/2} + \gamma_E + O(s - 1/2),$$

so the apparent pole is removable and

$$\mathcal{B}_h(1/2) = \gamma_E L_h(1/2) = \gamma_E M_h^2.$$

The remaining poles are

$$s = \rho - \frac{1}{2}$$

for nontrivial zeros, with residue

$$m_\rho L_h \left(\rho - \frac{1}{2} \right),$$

and

$$s = -2n - \frac{1}{2}, \quad n \geq 1,$$

for trivial zeros, with residue

$$L_h \left(-2n - \frac{1}{2} \right).$$

A multiple zero changes the residue by its multiplicity; it does not create a derivative term.

4.3 Contour displacement

For $c > 1/2$, Fourier inversion on the line $\Re s = c$ gives

$$J_h(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} e^{st} L_h(s) \left\{ \frac{\zeta'}{\zeta} \left(s + \frac{1}{2} \right) + \frac{1}{s - \frac{1}{2}} \right\} ds.$$

The integral is absolutely convergent because, uniformly in every fixed vertical strip,

$$L_h(\sigma + i\tau) = O_N((1 + |\tau|)^{-N})$$

for every N . Choose heights avoiding zero ordinates by $\gg 1/\log T$; on those horizontal segments $\zeta'/\zeta = O(\log^2 T)$, so the horizontal integrals vanish.

Move the left side to $\Re s = -2N - 1$. Repeated integration by parts and $\text{supp } K_h = [-1, 1]$ give, for $M \geq 3$,

$$|L_h(-2N - 1 + i\tau)| \leq C_M e^{2N+1} (1 + N)^M (1 + |\tau|)^{-M}.$$

The remaining vertical integral is consequently

$$O_{h,M} \left(e^{-(2N+1)(t-1)} (1 + N)^M \log(N + 2) \right),$$

which tends to zero when $t > 1$. Summing the crossed residues yields

$$\begin{aligned} J_h(t) &= \sum_{\rho} m_{\rho} L_h \left(\rho - \frac{1}{2} \right) e^{(\rho-1/2)t} \\ &\quad + \sum_{n=1}^{\infty} L_h \left(-2n - \frac{1}{2} \right) e^{(-2n-1/2)t}. \end{aligned} \tag{15}$$

Since L_h is even, $t = 2y$ proves (1).

The trivial-zero sum is absolutely convergent for $t > 1$. The zero-counting bound $N_{\zeta}(T) = O(T \log T)$, combined with rapid vertical decay of L_h , proves absolute and locally uniform convergence of the nontrivial zero sum after any fixed number of real derivatives.

4.4 Archimedean cancellation

For the reflected cross correlation, $C(0) = 0$, $C(r) = 0$ for $r \geq 0$, and $C(-r) = K_h(2y - r)$. Equation (5) gives

$$A_h^{\infty}(y) = - \int_0^{\infty} \frac{e^{r/2}}{e^r - e^{-r}} K_h(2y - r) dr.$$

Using

$$\frac{e^{r/2}}{e^r - e^{-r}} = \sum_{n=0}^{\infty} e^{-(2n+1/2)r},$$

Tonelli's theorem and $u = 2y - r$ give

$$A_h^{\infty}(y) = - \sum_{n=0}^{\infty} L_h \left(2n + \frac{1}{2} \right) e^{-(4n+1)y}.$$

The $n = 0$ term is $-M_h^2 e^{-y}$; the terms $n \geq 1$ cancel the complete trivial-zero series in (1). Combining with (13) proves the zero-only formulas (2) and (3).

5 Visible zero coefficients and infinite oscillation

5.1 Coefficient visibility

From (6), the n -th factor vanishes precisely at

$$w = \pi i \ell 2^{n+1}, \quad \ell \in \mathbb{Z} \setminus \{0\}.$$

The locally uniform nonvanishing of the tail away from factor zeros proves (7). Thus every off-critical-line zero is visible. A critical-line zero is invisible only if its ordinate is in

$$4\pi\mathbb{Z} \setminus \{0\}.$$

Conrey's Theorem 1 in [3] proves that a positive proportion of all nontrivial zeros are simple and lie on the critical line. Hence there are $\gg T \log T$ distinct simple critical-line zeros below height T . The invisible lattice has only $O(T)$ ordinates in that range, and a simple zero occupies at most one such ordinate. It follows unconditionally that infinitely many critical-line zeros are visible for the frozen packet.

5.2 A Laplace-transform Landau lemma

Lemma 5.1 (Landau). *Let $T_0 \geq 0$, and let $f : [T_0, \infty) \rightarrow [0, \infty)$ be locally integrable. Suppose its Laplace integral converges on a nonempty right half-line, and define*

$$F(s) = \int_{T_0}^{\infty} f(t)e^{-st} dt,$$

$$\sigma_c = \inf \left\{ \sigma \in \mathbb{R} : \int_{T_0}^{\infty} f(t)e^{-\sigma t} dt < \infty \right\}.$$

If σ_c is finite, then F is singular at the real point $s = \sigma_c$.

Proof. Assume that F is holomorphic near σ_c . Choose $\sigma_1 > \sigma_c$ and $r > 0$ so that $\sigma_1 - r < \sigma_c$ while the Taylor disk from σ_1 to $\sigma_1 - r$ remains inside the union of this neighborhood and $\Re s > \sigma_c$. For a small $\varepsilon > 0$ with $\sigma_1 - \varepsilon > \sigma_c$, the inequality $t^n \leq C_{n,\varepsilon} e^{\varepsilon t}$ justifies differentiation:

$$(-1)^n F^{(n)}(\sigma_1) = \int_{T_0}^{\infty} t^n f(t) e^{-\sigma_1 t} dt \geq 0.$$

Taylor's theorem and monotone convergence give

$$\begin{aligned} F(\sigma_1 - r) &= \sum_{n=0}^{\infty} \frac{(-r)^n F^{(n)}(\sigma_1)}{n!} \\ &= \int_{T_0}^{\infty} f(t) e^{-\sigma_1 t} \sum_{n=0}^{\infty} \frac{(rt)^n}{n!} dt \\ &= \int_{T_0}^{\infty} f(t) e^{-(\sigma_1 - r)t} dt < \infty, \end{aligned}$$

contradicting the definition of σ_c . □

5.3 Oscillation of the arithmetic interaction

The direct prime-power window gives the elementary bound

$$|\mathcal{I}_h(t/2)| = O_h((1+t)e^{t/2}), \quad t > 1.$$

Indeed, one may use $\sum_{n \leq X} \Lambda(n) \leq X \log X$. Thus the tail transform has a finite abscissa of convergence no greater than $1/2$.

At each visible critical-line zero $\rho = 1/2 + i\gamma$, the meromorphic continuation (14) has a genuine pole at $s = i\gamma$. Suppose $\mathcal{I}_h(t/2)$ were eventually nonnegative. Discarding the finite initial interval changes its transform by an entire function. The visible pole rules out an abscissa below 0, while absolute convergence gives

$$0 \leq \sigma_c \leq \frac{1}{2}.$$

But (14) is holomorphic at every real point of this interval: the pole at $1/2$ cancels, and $\zeta(u) \neq 0$ for real $1/2 \leq u < 1$. Lemma 5.1 gives a contradiction. Applying the same argument to $-\mathcal{I}_h(t/2)$ excludes eventual nonpositivity.

5.4 Oscillation of the complete splitting

Put $\mathcal{Q}_h(t) = \mathfrak{q}(h_{t/2}^R, h_{t/2}^L)$. The zero-only formula gives

$$\mathcal{Q}_h(t) = \sum_{\rho} a_{\rho} e^{(\rho-1/2)t}, \quad a_{\rho} = m_{\rho} L_h \left(\rho - \frac{1}{2} \right).$$

For every $T_0 > 1$,

$$\int_{T_0}^{\infty} \mathcal{Q}_h(t) e^{-st} dt = \sum_{\rho} \frac{a_{\rho} e^{(\rho-1/2-s)T_0}}{s - (\rho - 1/2)}.$$

Rapid coefficient decay makes the right side normally convergent away from its displayed poles. It has genuine nonreal poles at every visible critical zero and no real poles. The same Landau argument, applied separately to $\mathcal{Q}_h$ and to $-\mathcal{Q}_h$, excludes either eventual sign. Equation (8) transfers the conclusion to Δ_h .

Both $\mathcal{I}_h$ and Δ_h are continuous. Alternating positive and negative points can be chosen beyond increasing bounds, producing infinitely many disjoint sign-crossing intervals. This does not assert that the intervening zeros are simple or transverse.

6 Numerical diagnostics and finite certificates

No numerical statement in this section is used in Theorem 1.1. The analytic nonvanishing input comes from the exact zero set (7) and Conrey's theorem, not from computed zeta zeros.

6.1 Implementation A: direct prime-power side

The first implementation evaluates finite dyadic convolutions by an exact nonuniform B-spline identity. It enumerates every p^m in

$$|2y - \log(p^m)| < 1$$

and records the triple (p, m, p^m) and the exact coefficient $(\log p)/\sqrt{p^m}$. Packet truncations $N = 12, 14, 16$ are retained separately. An exact-rational computation certifies

$$0.063637725597 < \mathcal{I}_h(1/2) < 0.088006928482.$$

The finite window at $y = 1/2$ contains exactly

$$2, 3, 4, 5, 7.$$

The certificate includes the omitted infinite-convolution tail and does not replace the infinite oscillation proof.

6.2 Implementation B: independent Fourier and zero sides

The second implementation computes K_h by Fourier inversion of the sinc product on its direct side. Independently, it evaluates the truncated explicit formula through the product (6) and zeta zeros. At $y = 1.25, 1.75, 2.25$, the direct calculations retain respectively 15, 26, 50 prime powers. The two implementations agree to between 5.2×10^{-13} and 1.4×10^{-11} , inside their declared diagnostic budgets.

At 220-bit Arb precision, selected coefficients at zero indices 1, 2, 4 are certified to have strictly positive real intervals after enclosing the infinite product tail. The omitted

y	retained prime powers	$ \mathcal{I}_h^{(A)} - \mathcal{I}_h^{(B)} $
1.25	15	5.13×10^{-13}
1.75	26	8.25×10^{-12}
2.25	50	1.32×10^{-11}

Table 1: Cross-implementation diagnostics.

nontrivial-zero sum is not certified; therefore this computation makes no eventual-sign claim.

Machine-readable JSON and CSV files in the repository contain every retained prime power, including composite prime powers, rather than a primes-only projection.

7 Logical scope and independent audit

The analytic chain was reconstructed independently multiple times and subjected to separate circularity audits. The audited dependencies are:

- the frozen semilocal normalization (4);
- elementary properties of the fixed infinite convolution;
- absolute convergence of the Euler product in its initial half-plane;
- standard meromorphic continuation and the functional equation of ζ ;
- the zero-counting estimate $N_\zeta(T) = O(T \log T)$;
- Conrey’s unconditional positive-proportion theorem [3]; and
- the Landau lemma proved in this paper.

The proof does not assume:

- the Riemann hypothesis;
- global simplicity of zeta zeros;
- linear independence of zero ordinates;
- Weil positivity;
- an eventual sign;
- a numerical zero table;
- convergence of a determinant;
- completeness of any spectral realization; or
- information about semilocal ground states.

The exact coefficient transform was fixed before any zero theorem was applied. Its invisible lattice was derived from the packet alone, so the visible poles are not inserted by normalization. The two numerical implementations can be deleted without changing any step of the proof.

Corollary 7.1 (Fixed-packet no eventual parity). *For the frozen packet,*

$$\Delta_h(y) \leq 0 \quad \text{for all sufficiently large } y$$

and

$$\Delta_h(y) \geq 0 \quad \text{for all sufficiently large } y$$

are both false.

This corollary is deliberately not a theorem about the parity of a true ground state. Such a statement would require uniform estimates over a packet class and a variational comparison controlling the magnitude of the interaction. Neither is claimed here.

References

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